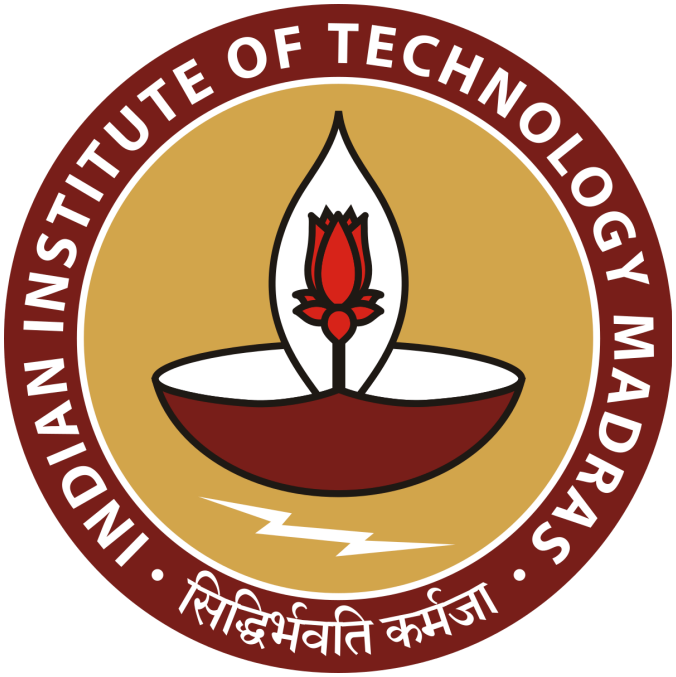
Multi-Dimensional Business Ecosystem Analysis of Tata Motors

A Mid-term report for the BDM capstone Project

Submitted by

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1 Executive summary

The core objective of the project is to analyze Tata Motor's performance over multiple years and across different areas using public data from their official reports, such as annual and quarterly financial statements. The analysis will look at key aspects like revenue growth, profitability, financial stability, and operational health. A significant outcome of this project will be the creation of a custom "Ecosystem Resilience Index (ERI)," which will be a single score that measures Tata Motor's ability to handle market disruptions and changes in business conditions.

This index will combine financial, operational, geographic, and supply chain indicators to systematically assess the company's adaptability over time and identify hidden patterns. The goal is to provide data-driven insights that can help with strategic decisions in a competitive and uncertain market.

Another key focus of the project is to analyze multi-year patterns in the company's performance to aid in strategic planning. This involves looking at financial results, revenue from different regions, and changes related to the supply chain. The project intends to uncover long-term trends that can help with making proactive decisions, improving operational flexibility, and strengthening strategic plans in a constantly changing market environment. The problem-solving approach includes extracting and compiling data from Tata Motor's reports, creating the ERI by selecting and standardizing metrics, and visualizing these trends with clear and easy-to-understand graphs.

2 Proof of Originality

The data presented and analyzed in this report is sourced exclusively from publicly available, official documents published by Tata Motors Limited. This includes annual reports, Business Responsibility and Sustainability Reports (BRSR), and quarterly financial statements, which are all accessible through the investors section of the company's official website. The source documents can also be accessed through the below link:

https://drive.google.com/drive/folders/1EKZHZqm7em33oN68GIaa0E2U5kupmDWD?usp=sharing

3 Metadata & Descriptive Statistics

3.1 Metadata

- **Dataset Name:** Tata Motors Multi-Dimensional Performance Data

Table1

Indicator	2015-16	2016-17	2017-18	2018-19	2019-20	2020-21	2021-22	2022-23	2023-24	2024-25
Revenue (₹ Cr)	278546	275247	294243	301938	261068	249795	278454	345969	437928	439695
EBITDA (₹ Cr)	2740	36912	31121	25765	12851	27249	28932	37042	62798	57649
PBT (₹ Cr)	13981	9315	11155	31371	10579	10474	7003	993	28932	34330
PAT (₹ Cr)	11024	7454	8988	28826	12070	13451	11441	2728	3057	28149
Net Cash from Operations (₹ Cr)	37900	30199	23857	18890	26632	29001	11441	35388	2414	63102
EBITDA Margin (%)	14.6	13.41	10.6	8.5	8.4	12.2	10.4	10.7	14.3	13.1
PAT to Sales (%)	3.9	2.7	3	9.5	4.6	3.5	4.1	0.7	7.1	6.2
Free Cash Flow (₹ Cr)	6543	268	1339	1539	9205	5317	9472	7840	26925	22348
Operating Profit	41776	36912	36872	29794	23914	35781	16298	16568	70568.8	65848
EPS (₹)	34.25	21.94	26.46	84.89	35.63	34.88	29.88	6.29	81.95	65.01
Current Assets (₹ Cr)	115315	116832	135972	123431	117046	122669	114639	150701	167718.24	159811
Total Assets (₹ Cr)	269298	273754	331350	307194	286168	275959	303799	338166	370664	378642
Current Liabilities (₹ Cr)	110820	115629	143219	145457	143024	128458	129033	155027	173617	166684
Total equity and liabilities (₹ Cr)	269298	273754	331350	246491	242883	233391	240030	336081	370663	378642
Total Debt (₹ Cr)	69360	78604	88950	106175	102897	99850	88478	125660	98500	62499
Net Debt/Equity Ratio	0.25	0.47	0.42	0.47	1.88	0.69	0.65	0.84	0.46	0.26
Capital Expenditure (₹ Cr)	32024	31751	42672	36635	31189	27049	29914	30977	33858	40531
Spend on R&D - TML (₹ Cr)	2217	2100	2398	2965	1324	1692	2139	2398	3926	3938

- **Dataset Link:** https://docs.google.com/spreadsheets/d/1PZMPE3osAxjHKZAHPOkZ9QVb_fAPfx26npTHxGZnC9Y/edit?usp=sharing

- **Description:** The dataset contains a wide range of metrics related to Tata Motors' business performance. It includes financial indicators (such as Revenue, EBITDA, PAT), operational data (such as vehicle sales, market share), and ESG/sustainability metrics (such as energy consumption, GHG emissions, and employee data). The data is structured to enable a multi-year analysis of the company's performance.

- **Source:** Publicly available official sources, including annual reports, BRSR (Business Responsibility and Sustainability Report) and quarterly financial statements.

- **Time-Period:** From the fiscal year 2015-16 to 2024-25.

- **Data Format:** The data is in a tabular format with the following columns:

    - Indicator: Name of the metric or key performance indicator.

    - 2015-16 to 2024-25: The values for each indicator for the corresponding fiscal year.

    - A detailed description of each performance indicator, including its definition, has been compiled in a supporting sheet. The sheet can be accessed here:

https://docs.google.com/spreadsheets/d/1PZMPE3osAxjHKZAHPOkZ9QVb_fAPfx26npTHxGZnC9Y/edit?usp=sharing

- **Purpose**: The primary aim of this dataset is to conduct a comprehensive, multi-year, and multi-dimensional analysis of Tata Motors. This analysis will evaluate the company's performance across key dimensions, including revenue growth, profitability trends, financial resilience, operational health, and geographic performance. The data will be used to develop a custom Ecosystem Resilience Index (ERI), a composite indicator that quantifies Tata Motors' capacity to withstand market disruptions and adapt to evolving business conditions.

- **Potential Insights**

    - The data will create an index to assess Tata Motors' business resilience using financial, operational, and supply chain indicators.
    - Analyzing multi-year patterns in financial metrics will provide a timeline-based understanding of the company's health.
    - The data will evaluate regional revenue and vehicle segment performance to highlight diversification and growth.
    - Cyclical patterns from the data will be used to understand demand fluctuations for strategic planning.

- **Usage:** This dataset is designed to provide a holistic view of Tata Motors' business ecosystem. The analysis will aid in making informed decisions related to strategic planning, operational improvements, and long-term resilience, thereby enhancing the company's overall business performance.

3.2 Descriptive Statistics

- A preliminary analysis of the dataset was conducted to understand its structure and key characteristics. This involved using fundamental Python commands to inspect the data's dimensions and contents.

- The initial inspection using df.shape reveals a DataFrame with 46 entries and 11 columns, including the Indicator column and 10 fiscal years from 2015-16 to 2024-25.

- The df.info() output provides a concise summary of the DataFrame, including column names, the number of non-null entries, and their data types.

```
[5] df.shape
```

```
(46, 10)
```

```
print("\n--- DataFrame Information ---")
df.info()
```

```
--- DataFrame Information ---
<class 'pandas.core.frame.DataFrame'>
Index: 46 entries, Revenue (₹ Cr) to C.I.F. value of imports - Raw material and components
Data columns (total 10 columns):
 #   Column   Non-Null Count  Dtype
---  ------   --------------  -----
 0   2015-16  46 non-null     float64
 1   2016-17  46 non-null     float64
 2   2017-18  46 non-null     float64
 3   2018-19  46 non-null     float64
 4   2019-20  46 non-null     float64
 5   2020-21  46 non-null     float64
 6   2021-22  46 non-null     float64
 7   2022-23  46 non-null     float64
 8   2023-24  46 non-null     float64
 9   2024-25  46 non-null     float64
dtypes: float64(10)
memory usage: 4.0+ KB
```

- The describe() command provides a statistical summary for the numerical data across all fiscal years, offering insights into the central tendency and spread of key metrics like Revenue, EBITDA, and Vehicle Sales.

```
--- Summary Statistics (Financial Indicators) ---
Indicator  Revenue (₹ Cr)  EBITDA (₹ Cr)    PAT (₹ Cr)
count           10.000000       10.00000     10.000000
mean        316288.300000    32305.90000  12718.800000
std          69609.422083    18102.99563   9054.149408
min         249795.000000     2740.00000   2728.000000
25%         276048.750000    26136.00000   7837.500000
50%         286394.500000    30026.50000  11232.500000
75%         334961.250000    37009.50000  13105.750000
max         439695.000000    62798.00000  28826.000000

--- Summary Statistics (Operational Indicators) ---
Indicator  Vehicle Sales (Total units)  Capital Expenditure (₹ Cr)
count                    1.000000e+01                   10.000000
mean                     1.186153e+06                33660.000000
std                      1.380067e+05                 4883.982414
min                      9.614630e+05                27049.000000
25%                      1.080227e+06                31030.000000
50%                      1.189466e+06                31887.500000
75%                      1.291712e+06                35940.750000
max                      1.380315e+06                42672.000000
```

The provided summary statistics offer a clear overview of Tata Motors' financial and operational performance over a 10-year period. The data reveals a strong upward trend in both revenue and profitability, as indicated by the mean values and the significant difference between the minimum and maximum values for each metric.

Financial Indicators

The financial statistics show a company that has experienced substantial growth while maintaining profitability. The mean Revenue of ₹316,288.30 Cr. and a maximum of ₹439,695.00 Cr. highlight a solid top-line performance. The EBITDA (a key measure of operational profitability) has a mean of ₹32,305.90 Cr. and a very wide range, with a minimum of just ₹2,740.00 Cr. and a maximum of ₹62,798.00 Cr. This wide standard deviation of ₹18,102.99 Cr. suggests significant volatility in core profitability, likely due to economic cycles or one-time events. The PAT (Profit After Tax) shows a similar pattern, with a mean of ₹12,718.80 Cr. and a large standard deviation, indicating that while the company is generally profitable, its bottom line has been inconsistent. The minimum PAT of ₹2,728.00 Cr. and maximum of ₹28,826.00 Cr. further illustrate this volatility.

Operational Indicators

The operational data reinforces the narrative of growth but also reveals a steady, upward trend. The mean Vehicle Sales of approximately 1.19 million units shows a strong market presence, with a clear positive trend from the minimum of 961,463 units to the maximum of 1.38 million units. This growth is supported by consistent investment, as indicated by the Capital Expenditure data. The mean CapEx of ₹33,660.00 Cr. and a tight range between the minimum of ₹27,049.00 Cr. and maximum of ₹42,672.00 Cr. suggest that Tata Motors has consistently and strategically invested in its operations, which is crucial for maintaining a competitive edge and supporting future sales growth.

4 Detailed Explanation of Analysis Process

4.1 Data Preparation, Organization, and Descriptive Statistics

The initial phase focused on ensuring the dataset was clean, consistent, and well-understood before any deeper analysis was performed.

- **Step 1: Data Integration and Cleaning** The data, sourced from Tata Motors' public disclosures, was first consolidated into a single structured dataset.
- **Step 2: Data Organization and Formatting** The dataset was organized with Indicator as the index and fiscal years as columns, a format that facilitates time-series analysis. All values were formatted to a numeric data type to enable mathematical operations.
- **Step 3: Descriptive Statistics** Commands such as df.info(), df.describe(), and df.shape were used to inspect the dataset's structure, dimensions, and statistical summary. This provided a foundational understanding of the data, including its range, central tendency, and non-null counts.

Justification: This methodical approach to data preparation is crucial for data integrity. By cleaning and organizing the data upfront, we eliminate errors and inconsistencies that could skew the results of later, more complex analyses. Descriptive statistics serve as a diagnostic tool, providing a necessary overview of the data's quality and characteristics.

4.2 Multi-Year Performance Trend Analysis

Financial Performance Visualization

- **Objective:** To visually represent the trends of key financial metrics over time.
- **Outcome Relevance:** This visualization contributes to the Multi-Year Performance Trends outcome by highlighting patterns in core financial metrics, supporting a clear understanding of Tata Motors' business health over time.
- **Methodology:** A dual-axis chart was created to combine revenue as a bar chart and EBITDA Margin as a line chart.
    - **Step 1:** The Revenue (₹ Cr) and EBITDA Margin (%) data were selected from the prepared dataset.
    - **Step 2:** A primary y-axis was set for Revenue (₹ Cr), and a bar chart was plotted to represent revenue for each fiscal year.
    - **Step 3:** A secondary y-axis was created on the right side of the chart for EBITDA Margin (%).
    - **Step 4:** A line chart was plotted on the secondary axis to display the EBITDA Margin trend, allowing for a clear comparison between top-line growth and operational profitability.
- **Justification:** The dual-axis chart is the most effective visualization for this purpose because it allows for the comparison of two metrics with vastly different scales (millions of rupees vs. a percentage) on a single, unified plot. This helps in understanding the relationship between the company's top-line growth and its operational efficiency.

Profitability Trends Visualization

- **Objective:** To show the year-on-year trends for core profitability indicators.
- **Outcome Relevance:** This visualization directly supports the Multi-Year Performance Trends goal of the Business Resilience Index by clearly showing the company's profitability evolution over time.
- **Methodology:** A grouped bar chart was used to compare EBITDA, PBT, and PAT side-by-side for each fiscal year.
    - **Step 1:** The EBITDA (₹ Cr), PBT (₹ Cr), and PAT (₹ Cr) data were selected from the dataset.

    - **Step 2:** A grouped bar chart was generated with fiscal years on the x-axis and each profitability metric represented by a separate bar within each year's group.
    - **Step 3:** Data labels were added on top of each bar to show the exact values, providing precise figures for analysis.
- **Justification:** A grouped bar chart is superior to a line chart for this data because it distinctly separates each profitability metric for a given year, making it easy to compare their relative values. It visually emphasizes the gaps and relationships between EBITDA, PBT, and PAT year by year.

Operational and Market Trend Visualization

- **Objective:** To combine sales volume and market share trends into a single, cohesive plot.
- **Outcome Relevance:** This visualization contributes to the Operational Efficiency and Market Insights outcome by combining key operational metrics and market position trends in a single, interpretable view.
- **Methodology:** A dual-axis chart was created with stacked bars for sales volume and lines for market share.
    - **Step 1:** Vehicles Sold - PV (units), Vehicles Sold - CV (units), Market Share - PV (%), and Market Share - CV (%) were selected.
    - **Step 2:** A primary y-axis was set for "Sales Volume (units)", and a stacked bar chart was plotted to show the breakdown of sales between passenger and commercial vehicles for each year.
    - **Step 3:** A secondary y-axis was created for "Market Share (%)".
    - **Step 4:** The Market Share - PV (%) and Market Share - CV (%) were plotted as separate lines on this secondary axis, allowing for a direct comparison of how the company's market position changes relative to its sales volume.
- **Justification:** This visualization method is highly effective as it allows for the simultaneous analysis of sales volume (an absolute number) and market share (a relative percentage). This helps in understanding whether a rise in sales volume is due to market growth or an increase in the company's market share.

Geographic Trend Visualization

- **Objective:** To display how the contribution of each geographic region to the company's total revenue has evolved over the decade.
- **Outcome Relevance:** This visualization supports the Geographic and Segment-Level Analysis by showing how regions and segments contribute to revenue and identifying growth and dependency areas.
- **Methodology:** A stacked bar chart was used to visualize the proportion of geographic revenue for each fiscal year.

    - **Step 1:** The Geographic Revenue indicators for all six regions were filtered from the dataset.
    - **Step 2:** A stacked bar chart was created with the fiscal years on the x-axis.
    - **Step 3:** For each year, the bars were stacked to show the percentage of revenue from each region, with a clear legend identifying each segment by color.

- **Justification:** A stacked bar chart is the most appropriate visualization for this data as it clearly shows the composition of total revenue by region for each year. This method makes it easy to identify shifts in revenue sources and track the growth or decline of specific markets over the long term.

4.3 Development of the Ecosystem Resilience Index Framework

The Ecosystem Resilience Index (ERI) is a core component of this project, designed as a composite indicator to provide a holistic, data-driven measure of Tata Motors' long-term resilience. The framework is built upon five key dimensions, each of which is evaluated using a selection of relevant performance indicators from the company's public disclosures.

Outcome Relevance: This ERI score directly achieves the goal of developing a Unified Evaluation Framework by combining multiple performance indicators into a single, clear measure of Tata Motors' resilience.

The ERI integrates metrics from five critical areas of business performance to provide a comprehensive score.

1. Financial Stability

This dimension assesses the company's ability to maintain financial health and flexibility during economic fluctuations.

Key Indicators:Revenue (₹ Cr), EBITDA (₹ Cr), Free Cash Flow (₹ Cr), Net Debt/Equity Ratio, PAT (₹ Cr)

2. Operational Health

This dimension evaluates the efficiency and strength of the company's core operations and market position.

Key Indicators:Vehicle Sales (Total units), Market Share - PV (%) & Market Share - CV (%), Inventory Turnover ratio, Employee Turnover (%)

3. Market and Geographic Diversity

This dimension assesses the company's exposure to different markets and its ability to mitigate risks through geographic diversification.

Key Indicators: Exports (units), Geographic Revenue percentages (India, China, USA, etc.), JLR Sold (units), Geographic Revenue - UK (%)

4. Strategic Investments and Innovation

This dimension measures the company's commitment to future growth, long-term capital formation, and innovation.

Key Indicators: Capital Expenditure (₹ Cr), Spend on R&D - TML (₹ Cr) & Spend on R&D - JLR (GB£ mn), Training Hours

5. Sustainability and Future Readiness

This dimension evaluates the company's non-financial performance, focusing on its environmental and social responsibility, which are increasingly critical for long-term resilience.

Key Indicators:GHG Emissions - Scope 1 & 2 (tCO2e), Energy Consumption - Direct (GJ), Lost Time Injury Frequency Rate (LTIFR), Total Employees

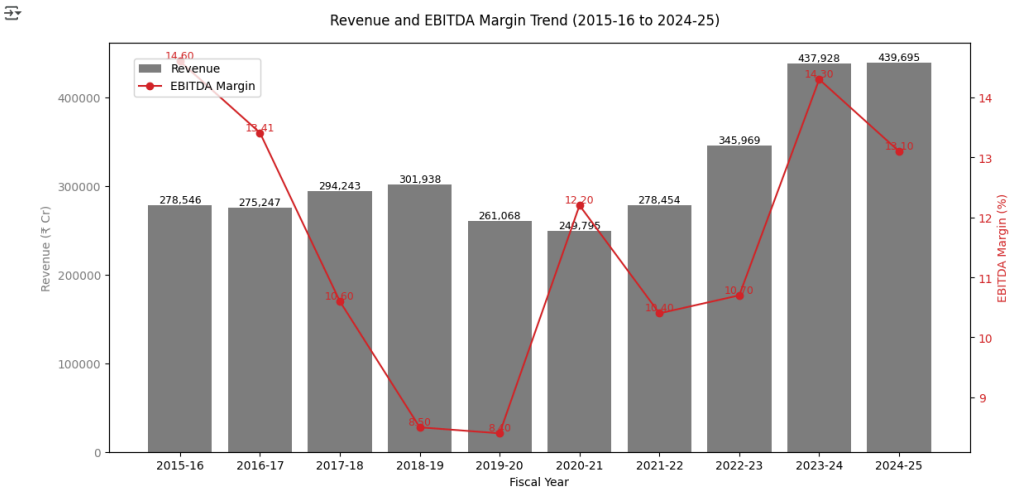
ERI Calculation Methodology

The ERI is calculated through a multi-step process:

1. **Metric Selection:** A curated set of indicators from each of the five dimensions is chosen to represent that dimension's overall health.
2. **Normalization:** All selected metrics are normalized to a common scale (0 to 1). This is done to ensure that all indicators, regardless of their unit of measurement (e.g., crores, percentages, units), contribute equally and fairly to the final score. For indicators where a lower value is better (e.g., Net Debt/Equity Ratio, GHG Emissions, Lost Time Injury Frequency Rate), an inverse normalization is applied.
3. **Weighting**: Weights are assigned to each of the five dimensions based on their strategic importance to the overall resilience of the company. These weights can be adjusted to reflect different analytical priorities.
4. **Aggregation:** The weighted and normalized scores for all indicators are summed up to produce a single ERI score for each fiscal year. This single score provides an overarching view of the company's resilience, allowing for easy monitoring of its performance over time.

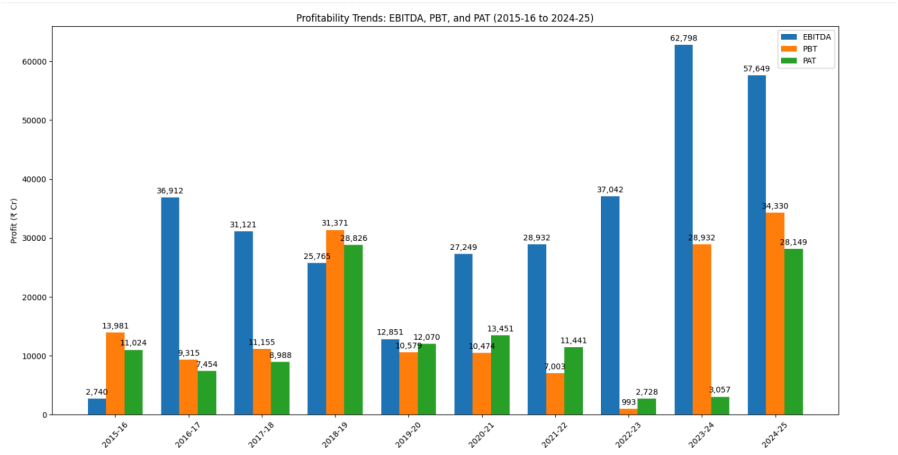
5 Results & Findings

Revenue and EBITDA Margin Trends



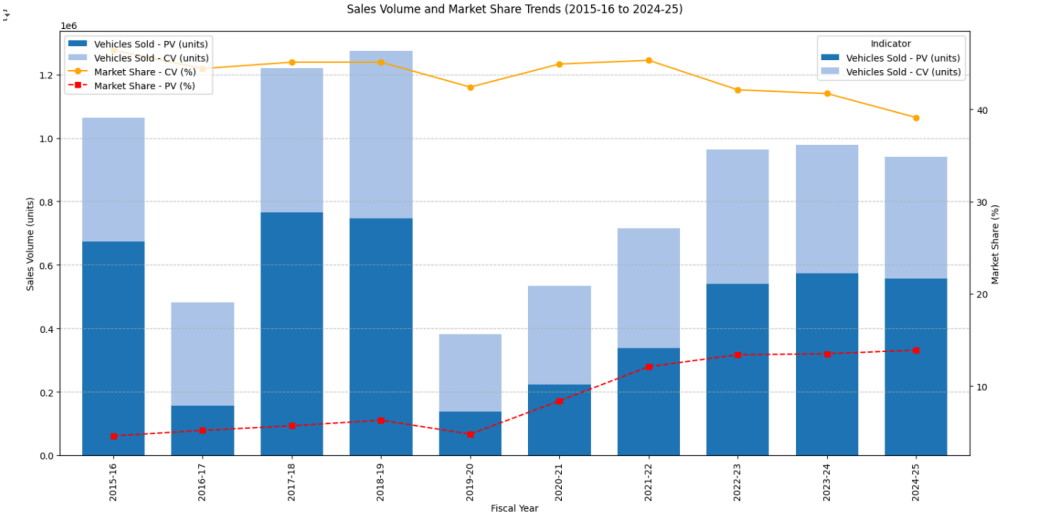
The dual-axis chart shows that revenue has grown significantly in recent years, indicating strong sales. However, the EBITDA Margin line has fluctuated, suggesting that profitability has not always kept pace with sales volume. This trend underscores the importance of operational cost management as the company scales.

Profitability Trends (EBITDA, PBT, and PAT)



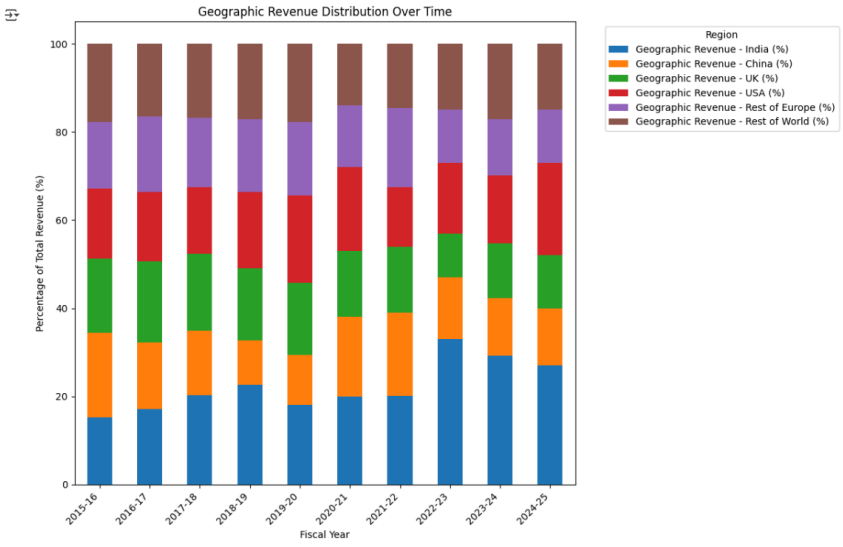
The grouped bar chart shows that while EBITDA has trended upward, the PBT and PAT are often disproportionately lower. This suggests that factors such as financial and tax obligations, rather than just core operations, are a significant influence on the company's final profitability.

Sales Volume and Market Share Dynamics



The dual-axis chart shows strong growth in total sales volume, primarily driven by the passenger vehicle (PV) segment. While PV market share has consistently risen, the commercial vehicle (CV) market share has declined, suggesting a strategic shift towards passenger vehicles and increased competition in the commercial sector.

Geographic Revenue Distribution



The stacked bar chart shows a significant shift in revenue concentration over the past decade. The contribution from India has grown substantially, while revenue from key international markets like China and the UK has been more volatile. This highlights a growing reliance on the domestic market for revenue and a shifting global footprint.